Zhao, Yuxuan

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Education

University of Illinois Urbana-Champaign | Urbana | IL USA — Aug 2023 – Present
PhD Candidate in Mathematics

ShanghaiTech University | Shanghai | China — Aug 2019 – Jul 2023
BS in Mathematics and Applied Mathematics

Research Experience

Doctoral Research | University of Illinois Urbana-Champaign — Dec 2024 – Present
Advisor: Prof. Daniel B. Cooney
Research Interests: Mathematical Biology, PDEs, Dynamical Systems

- Conduct research on nonlinear dynamical systems and PDE models using stability, bifurcation, and asymptotic analysis.
- Develop numerical simulations and computational pipelines in Python/MATLAB for parameter sweeps, pattern formation, and model validation.

Undergraduate Research | ShanghaiTech University — Dec 2022 – May 2023
Advisor: Prof. Shixiao Jiang
Thesis: *Some Meshless Methods on Smooth Manifolds*

- Implemented and reproduced meshless numerical methods for solving PDEs on smooth manifolds, including local RBF-FD and local MLS methods.

Publications and Manuscripts

- Finite–Wavelength Pattern Formation in a Nonlocal Self–Chemotaxis Model (Manuscript in preparation).
- Daecheol Kim, Yefei Zhang, **Yuxuan Zhao**, Daniel B. Cooney. The Keller–Segel Model for Chemotaxis on Metapopulation Graphs and Graphons. arXiv preprint, arXiv:2610.31049 (2026).
- Yefei Zhang, **Yuxuan Zhao**. Spatial Pattern Formation from Multi-Agent Learning in Public Goods Dilemmas. arXiv preprint, arXiv:2610.12321 (2026).
- **Yuxuan Zhao**, Kaisheng Zhu, Yefei Zhang, Daniel B. Cooney. Pattern Formation in a Spatial Public Goods Dilemma due to Diffusive or Directed Motion. *Bulletin of Mathematical Biology*, **88**, 173 (2026). doi:10.1007/s11538-026-01733-0.

Research Mentorship

IML Project Mentor (Volunteer) | University of Illinois Urbana-Champaign — Jan 2026 – May 2026
Urbana, IL USA

- Mentored undergraduate students interested in mathematical research through the Illinois Mathematics Lab (IML).
- Designed and supervised small-scale research projects suitable for undergraduate researchers.
- Worked under the guidance of faculty instructors, serving as the graduate mentor for student research teams.

Training Program

INMAS Training Program | University of Illinois Urbana-Champaign — Oct 2024 – May 2025
Urbana, IL USA

- Selected for competitive industrial math & statistics training program.
- Applied Python, statistics, and machine learning to interdisciplinary data-driven projects.
- Delivered technical presentations to peers and mentors.

Work Experience

Teaching Assistant and Grader | University of Illinois Urbana-Champaign — Aug 2023 – Present
Urbana, IL USA

- Calculus, differential equations, machine learning, real analysis, and topology.

PrairieLearn Summer Intern | PrairieLearn, Inc. — May 2025 – Aug 2025
Urbana, IL USA

- Developed Python-based interactive assessment and auto-grading systems using NumPy and SymPy.
- Collaborated with a cross-functional team on implementation and deployment.

Skills & Interests

Languages: Fluent in English and Mandarin Chinese

Technical Skills: Python (NumPy, Pandas, SciPy, SymPy); MATLAB; LaTeX; Git; Microsoft Suite

Interests: Hiking, table tennis